

Contents

1) Existing Studies	2
1.1) 3D printed metamaterial lenses for microwave antennas (FFI 2017)	2
1.1.1) Exploratory study	2
1.1.2) Methodology & Construction	2
1.1.3) Experimental Findings	2
1.2) Design of a metamaterial Luneburg lens antenna (CEI 2022)	2
1.2.1) Methodology & Construction	3
1.2.2) Experimental Findings	3
2) Manufacturing Observations	4
2.1) SLA (Stereolithography)	4
2.2) SLS (Selective Laser Sintering)	4
2.3) Manufacturing Pitfalls	4
2.4) FMD with multiple printheads	4
3) Construction Types	5
3.1) Luneburg Lenses	5
3.2) GRIN (Gradient Index) Lenses	5
3.3) Phase-Screens	5
3.4) Metamaterial based antennas	5
3.5) Perfectly matched Layer Radome	6

1) Existing Studies

1.1) 3D printed metamaterial lenses for microwave antennas (FFI 2017)

- **Authors:** Stein Kristoffersen, Karina Vieira Hoel
- **Source:** FFI-RAPPORT 17/00415
- **Research Context:** Exploratory study financed by the Norwegian Defence Research Establishment (FFI) to build competence in materials with spatially varying permittivity for military applications.
- **Filename:** 17-00415.pdf
- **Link:** www.ffi.no

1.1.1) Exploratory study

- Overview of existing use cases - Luneburg, GRIN lenses, brief mention of useage of metamaterials in creation of phase screen, antennas or radomes
- Most of the menioned papers were more theoretical and on the older side.

1.1.2) Methodology & Construction

- **Unit Cell Design:** The study utilized a 3.55 mm lattice grid structure built by unit cubes intersected in three spatial directions.
 - structure realized was a GRID lense, but varying density was achived using a 3D lattice grid strucure.
- **Material Selection:** Used PA2200 nylon, initially believed to have $\epsilon_r \approx 3.6$ (glass-mixed version), but measured at $\epsilon_r \approx 2.5 - 2.8$ for 100% density.
- **Simulations:** Performed using ANSYS HFSS, modeling ϵ_r vs. frequency for various cube sizes.

1.1.3) Experimental Findings

- **Test Case 1 (Lattice Blocks):** Achieved a span in measured ϵ_r from 1.25 to 2.45.
- **Test Case 2 (Flat Lens):** Demonstrated a frequency-independent gain increase of approximately 3 dB over the 8-18 GHz range.
- **Loss Characteristics:** Measurements with the lens at the antenna aperture showed very little loss, confirming the efficiency of the metamaterial approach.
- **3D printing**
 - Authors mentioned repeated failures when trying to use standard SLA 3D printeing to use normal SLA 3D printing and were forces to rely on SLS.
 - Even in SLS they had to use higher minimum thicnkess of 0.7 mm then desired.
- It's vital to measure permitvity of the material that has been manufactured using the same process as the final lens.
 - Measured permitivty will be smaller than that of material itself.
 - Note: authors haven't borthered with it but probably the measured strucutres should have same form as those used in the final lens.
- Authors reported significant problems in simulating the structure using ANSYS HFSS.

1.2) Design of a metamaterial Luneburg lens antenna (CEI 2022)

- **Authors:** Jiaxuan Yue, Shunlian Chai, Chengmian Zhou, Ke Xiao
- **Source:** 2022 2nd International Conference on Computer Science, Electronic Information Engineering and Intelligent Control Technology (CEI)
- **Research Context:** Demonstration of a 50 mm in radius Luneburg lense created using SLA pritning, achived an improvement in gain of 7.41 dB
- **Filename:** Design_of_a_metamaterial_Luneburg_lens_antenna_based_on_3D_printing_technology.pdf

- **Link:** www.ieeeexplore.ieee.org

1.2.1) Methodology & Construction

- **Unit Cell Design:** Each unit consists of a variable-sized dielectric cube in the center with three connecting rods (0.8 mm fixed width) parallel to the X, Y, and Z axes.
 - Unit period of 5 mm was chosen to be significantly smaller than the center wavelength of the X-band (10 GHz).
 - Lens was fed from a standard WR-90 waveguide (Just open port without any antenna).
- **Parameter Retrieval:** Used the S-parameter retrieve method proposed by D. R. Smith (2005) to extract equivalent permittivity.
- **Manufacturing:** Produced a 50 mm radius lens using Stereo Lithography Apparatus (SLA) with C-UV 9400E photosensitive resin ($\epsilon_r \approx 3.2 - 4.0$).

1.2.2) Experimental Findings

- **Gain Improvement:** The antenna gain at 10 GHz was 14.98 dB, a 7.41 dB increase compared to a single waveguide feed.
- **Calculation of parameters:** Authors mention a method of calculating permittivity of these period structures
 - First S parameters of individual cube are acquired using EM field simulator.
 - Then using formula

$$n = \pm \frac{1}{kd} \left\{ \arccos \left[\frac{1 - S_{11}^2 + S_{21}^2}{2S_{21}} \right] + 2\pi m \right\}$$

$$z = \pm \sqrt{\frac{(1 + S_{11})^2 - S_{21}^2}{(1 - S_{11})^2 - S_{21}^2}}$$

$$\epsilon = \frac{n}{z}$$

- Formula described in: D. R. Smith, D. C. Vier, T. Koschny, and C. M. Soukoulis, “Electromagnetic parameter retrieval from inhomogeneous metamaterials,” *Phys. Rev. E.*, vol. 71, no. 3, p. 036617, 03/22/ 2005
- No problems in regards to simulations or manufacturing using SLA were mentioned in the study
- **Beamwidth:** Main lobe width narrowed from 28.26° at 8 GHz to 18.84° at 12 GHz.
- **Scanning Capability:** The lens rotated 45° with basically unchanged pattern and gain, proving “good spatial dynamic scanning ability”.

2) Manufacturing Observations

- The effective relative permittivity (ϵ_r) depends on the material density in a systematic and predictable way, provided the resolution is significantly higher than the wavelength.
- **Note:** need to figure out how to design these structures - using standard slicers will not cut it, there cannot be any support and we cannot rely on classic infill algorithms - whole structure needs to be described in STL file. Unless some STL supports some “density” parameter that the slicers would be able to understand.

2.1) SLA (Stereolithography)

- **Contradictory Results:** FFI found SLA resolution “disappointing” for fine grids, whereas CEI 2022 successfully used it for a 50mm sphere

2.2) SLS (Selective Laser Sintering)

- **Efficiency:** Required for high-quality grids in the FFI study, allowing for one-print production of complex objects.
- **Note:** Expensive to use, would need to verify if even is available in NTUST

2.3) Manufacturing Pitfalls

- **Density Achievement:** Printing often fails to reach 100% density, shifting measured ϵ_r lower than theoretical values.
- **Maintenance:** Removing excess powder from the internal grid of larger objects is a significant logistical hurdle.

2.4) FMD with multiple printheads

- Haven’t found a study that uses it
- **Note:** NTUST has a whole research group dedicated to this

3) Construction Types

3.1) Luneburg Lenses

- **Mechanism:** Spherical structure with decreasing refractive index with higher distance from the center.
- **Design Rule:** Theoretically follows

$$n(r)^2 = \varepsilon_{r(r)} = 2 - \left(\frac{r}{R}\right)^2,$$

where n is the refraction index, ε_r the permittivity, and R is the total radius of lense.

- Using normal manufacturing methods creating of Luneburg Lenses is exceedingly difficult.
- Strucure is creating using an uniform latice with cubes being placed in intersection points, with increasing dimensions of cubes permittivity increases.
- **Mentioned in:**
 - 3D printed metamaterial lenses for microwave antennas (FFI 2017)
 - Design of a metamaterial Luneburg lens antenna (CEI 2022)

3.2) GRIN (Gradient Index) Lenses

- **Mechanism:** Flat structures with radial permittivity gradients expressed as

$$n(r)^2 = \varepsilon_{r(r)} = \left(n_0 - \frac{\sqrt{L^2 + r^2} - L}{t} \right)^2,$$

where n is the refraction index, ε_r the permittivym, n_0 refractive index at 100 % infill, r distance from center, L the local distance, and t the lens thickness.

- **Note:** Really easy construction, definitely compatible with FDM process, might be usefull as a starting point.
- **Mentioned in:**
 - 3D printed metamaterial lenses for microwave antennas (FFI 2017)

3.3) Phase-Screens

- **Function:** Phase-Switched Screens (PSS) used for radar signature control and interference mitigation (e.g., wind turbine blades).
- surface is made up of electronically controllable or passive elemtns, that enable deflection in specific direction or absrob the energy
- **Mentioned in:**
 - passsing mention: 3D printed metamaterial lenses for microwave antennas (FFI 2017)
 - concept demonsration: B. Chambers, A. Tennant: "The phase switched screen", IEEE Antennas and Propagation Magazine, Vol 46, No 6 (2004)

3.4) Metamaterial based antennas

- **Function:** Enables construction of physically smaller antennas, or achieves beam steering using different mechanism then phase array atennas.
- **Mentioned in:**
 - passsing mention: 3D printed metamaterial lenses for microwave antennas (FFI 2017)
 - Y. Dong, T. Itoh: "Metamaterial-based antennas", Proceedings of the IEEE, Vol 100, No 7, (July 2012)

3.5) Perfectly matched Layer Radome

- **Function:** Creation of close to ideal radome, that doesn't refract EM waves, regardless of the incidence angle
- **Mentioned in:**
 - passing mention: 3D printed metamaterial lenses for microwave antennas (FFI 2017)
 - Schultz et al.: "Radome compensation using matched negative index or refraction materials" US patent no 6,788,273 B1, (September 2004)
 - N Michishita, Y. Yamada: "Metamaterial radome composed of negative refractive index lens for mobile base station antennas", Presented at IEEE ATC'14, (2014)